Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes

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Next-generation ground-based gravitational-wave (GW) detectors are expected to detect millions of binary black hole mergers during their operation period. A small fraction (∼0.1–1%) of them will be strongly lensed by intervening galaxies and clusters, producing multiple copies of the GW signals. The expected number of lensed events and the distribution of the time delay between lensed images will depend on the mass distribution of the lenses at different redshifts. Warm dark matter and fuzzy dark matter models predict lower abundances of low-mass dark matter halos as compared to the standard cold dark matter. This will result in a reduction in the number of strongly lensed GW events, especially with small time delays. Using the number of lensed events and the lensing time delay distribution, we will be able to put a lower bound on the mass of the warm and fuzzy dark matter particles from a catalog of lensed GW events. Our first forecasts suggest that the expected bounds from GW strong lensing from next-generation detectors are better than the current constraints.

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Introduction—The existence of dark matter (DM) is firmly established by a variety of observations. The possible list of DM candidates ranges from extremely light elementary particles [1,2] to supermassive primordial black holes (BHs) [3]. DM candidates can be classified according to their velocity dispersion, which defines a free streaming length scale. Below this length scale, all of the cosmological density perturbations are wiped out, so no structure can form below this scale. *Cold* DM (CDM), such as the weakly interacting massive particles [4], axions [5], and primordial BHs [6], has small free streaming length scales and does not affect the cosmological structure formation. On the other hand, *hot* DM such as neutrinos is highly relativistic. Free streaming of such relativistic particles would erase perturbations in the matter density even on the scale of galaxy clusters ($\sim 10^{15} M_\odot$). The very existence of such large-scale structures has ruled out hot DM [7]. In between there exists another class, called *warm* DM (WDM), such as gravitino [8] and sterile neutrino [9]. They are non-relativistic but have non-negligible velocity dispersion. The resulting shorter free streaming lengths can erase structures on galaxy scales. Thus the existence of galaxies can put some (rather weak) constraints on the mass of the WDM particle.

Despite decades of experimental effort, we have not been able to detect any CDM candidates so far. In addition, though predictions of the ΛCDM cosmological model match with observed large-scale structure, subgalactic observations might be in conflict with the CDM predictions. One is the apparent underabundance of satellites in the Milky Way, as compared to the earlier CDM simulations [10–13]. (However, with better simulations and more careful characterization of the observational selection effects, this discrepancy might be already resolved [14].) The second is the observed discrepancy between the inferred DM density profiles of low-mass galaxies and that predicted by CDM simulations [15]. Simulations typically predict "cuspy" profiles (steep density profiles at the center) while observations suggest the existence of "cores" (softer density profiles at the center).

While some of these apparent discrepancies between CDM models and observations could be attributed to astrophysical reasons (such as the effect of baryons), several new DM candidates have also been proposed to address them. WDM is the simplest departure from CDM, endowing the DM with a small velocity dispersion. A WDM particle with a mass in the keV range predicts the suppression of structures at small scales (∼100 kpc) without affecting the large scales (∼Mpc), thus explaining the missing satellites. Another model called *self-interacting DM* [16] adds a self-interaction cross section to the DM. The resulting elastic scattering between the DM particles in the inner galactic regions redistributes energy, producing the effect of a core. Fuzzy DM (FDM) particles are ultralight bosons (mass $\sim 10^{-22}$ eV), with de Broglie wavelength larger than the interparticle separation. The resulting wavelike behavior leads to the formation of solitonic cores at the center of haloes, and density granules on scales smaller than ~kpc are erased, while the large-scale structure is indistinguishable from CDM [17].

The observation of Lyman-α forest provides a lower limit $m_{\rm WDM} > 3$–5 keV on the WDM mass [18–21]. Combining this with strong gravitational lensing [22] and the abundance of Milky Way satellites has resulted in a limit $m_{\rm WDM} > 6$ keV [23]. Recent JWST observations of lensed quasars suggest $m_{\rm WDM} > 6.1$ keV [24]. Different cosmological datasets put upper bound $m_\psi \geq 10^{-22}$ eV on the mass of FDM [25–28]. Stronger constraints are obtained from the survival of an old star cluster in an ultrafaint dwarf galaxy Eridaus II ($m_\psi \gtrsim 10^{-19}$ eV) [29] and from the sizes and stellar kinematics of ultrafaint dwarf galaxies ($m_\psi \geq 3 \times 10^{-19}$ eV) [30].

Gravitational-wave (GW) observations offer new probes of DM (see, e.g., [31]). The presence of a DM overdensity surrounding a BH can have an impact on the GWs emitted by compact binary inspirals, which could be detectable by LISA [31–33]. Ultralight bosons, such as axions, can affect the mass and spin of BHs by forming gravitationally bound states around them [34–36]. GW emission from such objects could be detected [37,38]. FDM can also be indirectly detected using pulsar timing arrays if the oscillation frequency falls within their detection band [39].

Gravitational lensing of GWs offers yet another avenue for probing the nature of DM (see, e.g. [40–47]). The next-generation (XG) ground-based GW detectors will detect millions of binary BH mergers (BBH) out to high redshifts ($z \sim 10$–100) [48]. About 0.1–1% [49–57] of them will be strongly lensed by the galaxies and clusters hosted in these DM halos, producing multiple copies of the GW signals. The time delay between the lensed copies of these GW signals can be accurately measured. The exact fraction of lensed mergers and the distribution of lensing time delay will depend on the mass distribution of lenses at various redshifts [58] as well as the cosmological parameters [49]. In this Letter, we propose a statistical probe to constrain the mass of the WDM particle using a catalog of strongly lensed GW detections. If the DM is warm, this will hinder the formation of low-mass halos. This suppression in the abundance of low-mass halos will result in a reduction in the number of lensed events with small time delays as small time delays are mostly produced by low-mass lenses.

This approach is closely connected to our earlier work [49] on constraining the cosmological parameters from strongly lensed GW signals. Our method does not rely on accurate knowledge of the source location of the individual signals or the properties of the corresponding lenses. Indeed, the number of lensed events as well as the time delay distribution will also depend on the distribution of source properties (e.g., mass and redshift distribution of BBHs [56,59,60]) as well as the lens properties (e.g., the mass function of the DM halos [61] and the lens model [62,63]). If the distributions of the source and lens properties are known from other observations or theoretical models (e.g., from the observation of unlensed GW signals and cosmological simulations), then the mass of the WDM can be inferred from the observed number of lensed events and their time delay distribution.

We forecast that BBH observations during 10 yrs of operation of XG detectors [64,65] will provide constraints ($m_{\rm WDM}^{-1} < 0.04$–0.06 keV^{-1}) that are significantly better than the current constraints ($m_{\rm WDM} > 6$ keV). We simply translate the constraints on $m_{\rm WDM}$ to the mass of the FDM particle (m_ψ). An optimistic assumption of the merger rate will give us the constraint $m_\psi \gtrsim 7 \times 10^{-19}$ eV, which is an improvement over existing bounds ($m_\psi > 3 \times 10^{-19}$ eV).

Warm dark matter—Free streaming of WDM particles suppresses primordial perturbations at scales smaller than the *free streaming scale*. Fitting functions for modeling the WDM transfer function have been proposed in different studies [13,66,67]. They give us a prescription to convert the power spectrum $P_{\rm CDM}(k)$ of linear perturbations in the CDM model to the same in the WDM model [$P_{\rm WDM}(k)$]. We use the transfer function given in [67],

$$T(k) = \left[\frac{P_{\rm CDM}(k)}{P_{\rm WDM}(k)}\right]^{1/2} = [1 + (\alpha k)^{2\mu}]^{-5/\mu}, \qquad (1)$$

where $\mu = 1.12$ and

$$\alpha = 0.049\left(\frac{m_{\rm WDM}}{\rm keV}\right)^{-1.11}\left(\frac{\Omega_{\rm WDM}}{0.25}\right)^{0.11}\left(\frac{h}{0.7}\right)^{1.22}\ h^{-1}\,{\rm Mpc} \qquad (2)$$

is called the *effective free streaming scale*. Here, $\Omega_{\rm WDM}$ is the energy density in the form of WDM, and h is the Hubble constant in units of 100 km/s/Mpc. We can introduce another length scale, the *half mode length scale* $\lambda_{\rm hm}$, where the WDM transfer function becomes half: $\lambda_{\rm hm} = 2\pi\alpha(2^{\mu/5} - 1)^{-1/2\mu}$. This length scale introduces a *half mode mass scale* $M_{\rm hm} = (4\pi/3)\bar{\rho}(\lambda_{\rm hm}/2)^3$, where $\bar{\rho}$ is the averaged density of the halo. Abundances of DM haloes with mass below $M_{\rm hm}$ will be suppressed compared to CDM, while the masses above $M_{\rm hm}$ are unaffected.

We use different halo mass functions (HMFs) that model the comoving number density $dn_{\rm CDM}/dM$ of CDM halos in different mass ranges. Given a HMF in CDM, the same in the WDM model for a particular $m_{\rm WDM}$ is obtained using the fitting formula given in [68]:

$$\frac{dn_{\rm WDM}/dM}{dn_{\rm CDM}/dM} = \left(1 + \frac{M_{\rm hm}(m_{\rm WDM})}{M}\right)^{-\beta}, \qquad (3)$$

where $\beta = 1.16$. The dependence on $m_{\rm WDM}$ comes through $M_{\rm hm}$. To obtain HMF in the WDM model, we use the HMFCALC package [69]. For our main analysis, we consider

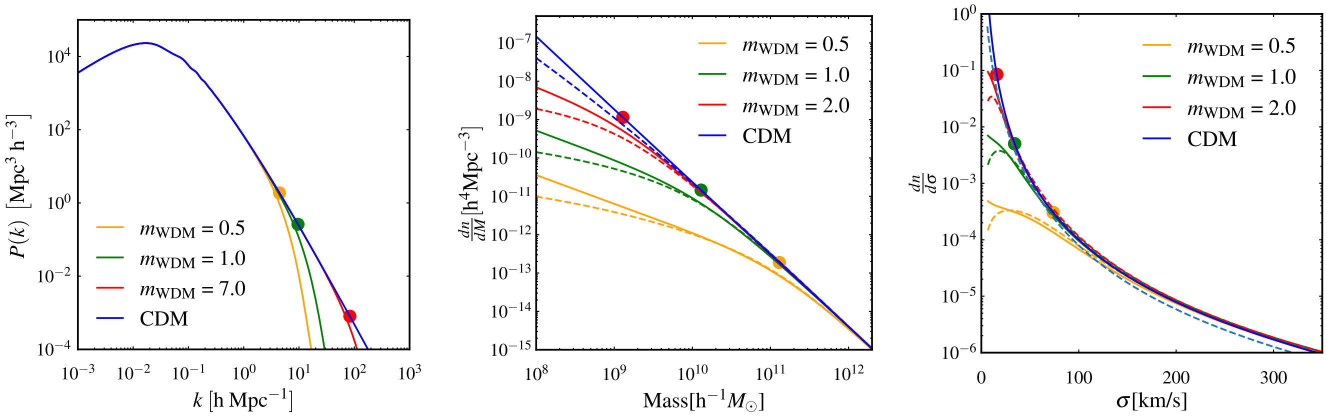

FIG. 1. Left panel: power spectrum of linear perturbations as predicted by the CDM model as well as the WDM model with different values of $m_{\rm WDM}$ (in keV). Half mode scales for different $m_{\rm WDM}$ are shown by the filled circles. Middle panel: HMF for CDM and WDM at redshift $z = 0$. Note the suppression in the number density of lower mass halos in the WDM model. Filled circles with different colors denote the half mode mass scale for different $m_{\rm WDM}$. Right panel: distribution of the velocity dispersion of lenses hosted by the CDM and WDM halos. Filled circles denote the velocity dispersion of the corresponding half mode mass. Note the suppression of low σ halos in WDM. In the middle and right panels, solid and dashed curves represent the Behroozi [70] and Jenkins [71] HMF models, respectively.

the Behroozi [70] model of the CDM HMF. In order to estimate the effect of using an incorrect CDM HMF model, we also consider the Jenkins [71] model of the same implemented in the same package.

These DM halos trap baryons and will eventually host galaxies, which act as gravitational lenses. We approximate these lenses as singular isothermal spheres (SISs) [72], parametrized by their dispersion velocity σ. Note that the lensing potential is sourced by DM and baryons and the total effect is reasonably well described by an SIS profile (see, e.g., [73]). To compute σ, we assume that the halos are spherically symmetric and virialized, with average density $\bar{\rho}$ and radius r:

$$\sigma \simeq \sqrt{\frac{GM}{r}}, \qquad M = \frac{4}{3}\pi r^3 \bar{\rho}. \tag{4}$$

Figure 1 shows the power spectrum, the HMF, and the σ distribution of lenses derived from the HMF, as predicted by the CDM model as well as the WDM model corresponding to different WDM masses. These can be used to compute the expected number of strongly lensed GW signals and the distribution of the time delay between the lensed copies of GW signals.

Expected constraints on warm DM—We first ask the following question: if the DM is actually cold, how well can we constrain $m_{\rm WDM}$ using future observations of lensed GWs? In order to answer this, we simulate a population of BBH mergers with redshift distribution given by [74]. We assume a nominal BBH detection rate $R = 5 \times 10^5$ yr^{-1} (see, e.g., [64,75]) and an observation period $T_{\rm obs} = 10$ yrs for XG detectors. We neglect the selection effects in detection as XG detectors are anticipated to detect all BBH mergers out to large redshifts ($z_s \sim 10$–100) [48,76]. Since the expected statistical errors (∼ms) in measuring the time of arrival of GW signals are well below the lensing time delays (∼ minutes to years), we also ignore these errors.

The mass distributions of the lenses at various redshifts are described by the CDM HMF model of [70], converted to the WDM HMF using Eq. (3). We consider DM halos in the mass range 10^8–$10^{15} M_\odot$. Lenses are modeled using the SIS model, using Eq. (4) for converting the halo mass M to the velocity dispersion σ of the lens. This allows us to compute the strong lensing optical depth $P_\ell(z_s|m_{\rm WDM})$ for sources located at different redshifts z_s. The optical depth is convolved with the redshift distribution $p_b(z_s)$ of the BBH mergers to compute the expected number Λ of lensed events:

$$\Lambda(m_{\rm WDM}) = RT_{\rm obs} \int_0^{z_s^{\max}} p_b(z_s) P_\ell(z_s|m_{\rm WDM}) dz_s. \tag{5}$$

Here, $z_s^{\max} \simeq 15$ is the redshift horizon of the detector.

In the SIS lens model, the time delay between the two images is (see, e.g. [72])

$$\Delta t = \frac{32\pi^2 y}{c} \left(\frac{\sigma}{c}\right)^4 (1 + z_\ell) \frac{D_\ell D_{\ell s}}{D_s}, \tag{6}$$

where z_ℓ is the lens redshift; y is the projected location of the source on the lens plane (in units of Einstein radius); and D_ℓ, D_s, and $D_{\ell s}$ are the angular diameter distances to the lens, to the source, and between the lens to the source, respectively. The mass of the WDM particle will also have an imprint on the lensing time delay distribution since the distribution of σ is a function of $m_{\rm WDM}$ [see Eqs. (3) and (4)]. We compute the expected time delay distribution

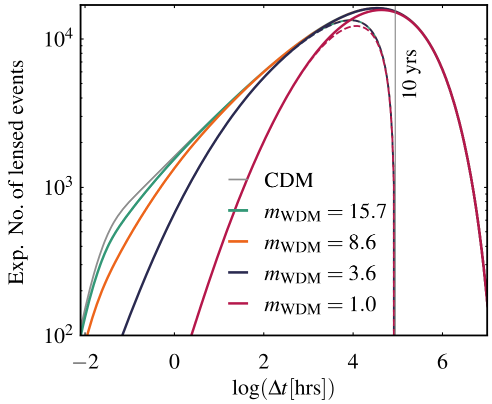

FIG. 2. Expected distributions of time delay between strongly lensed GW signals, corresponding to different values of $m_{\rm WDM}$. Note the suppression in the number of lensed events compared to CDM, especially for lower time delays, which is the reflection of the absence of the lower mass halos for smaller $m_{\rm WDM}$. Here, we assume the redshift distribution model of BBH mergers by [74] and a detection rate of $R = 5 \times 10^5\ \mathrm{yr}^{-1}$ by XG detectors. Solid curves show the intrinsic time delay distributions while dashed curves show the distributions measurable using an observation period of $T_{\rm obs} = 10$ yrs.

$p(\Delta t|m_{\rm WDM})$ by marginalizing the distribution of time delay over all other parameters $\vec{\lambda} \equiv \{y, \sigma, z_\ell, z_s\}$ that influence the time delay:

$$p(\Delta t|m_{\rm WDM}) = \int p(\Delta t|\vec{\lambda}, m_{\rm WDM}) p(\vec{\lambda}|m_{\rm WDM}) d\vec{\lambda}, \quad (7)$$

where $p(\vec{\lambda}|m_{\rm WDM})$ is the expected distribution of $\vec{\lambda}$, given $m_{\rm WDM}$ and the cosmological parameters (see Appendix A). We assume the following values of cosmological parameters: $\Omega_m = 0.316, H_0 = 67.3, \sigma_8 = 0.816$ [77].

Figure 2 shows the expected number of lensed events [i.e., $\Lambda(m_{\rm WDM}) p(\Delta t|m_{\rm WDM})$] as a function of the lensing time delay for different values of $m_{\rm WDM}$. For smaller $m_{\rm WDM}$, the absence of lower mass halos results in the suppression of lensed events with lower time delays. Using these differences in the time delay distribution and total number of lensed events, we will be able to either measure the mass of the WDM particle or put a lower bound on $m_{\rm WDM}$. In practice, we put an upper bound on $m_{\rm WDM}^{-1}$ since it has a convenient lower bound of zero (in the limit of CDM, $m_{\rm WDM} \gg$ keV). Note that both $\Lambda(m_{\rm WDM})$ and $p(\Delta t|m_{\rm WDM})$ are altered by the finite observing time $T_{\rm obs}$. The calculation of these quantities, $\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs})$ and $p(\Delta t|m_{\rm WDM}^{-1}, T_{\rm obs})$, are detailed in Appendix A.

To simulate an observing scenario with N detections of lensed events, we draw one sample of N from a Poisson distribution with mean $\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs})$. Further, we draw N samples of lensing time delay, $\{\Delta t_i\}_{i=1}^N$, from $p(\Delta t|m_{\rm WDM}^{-1} \simeq 0, T_{\rm obs})$. Using N and $\{\Delta t_i\}_{i=1}^N$ we evaluate the likelihoods $p(N|m_{\rm WDM}^{-1}, T_{\rm obs})$ and $p(\{\Delta t_i\}|m_{\rm WDM}^{-1}, T_{\rm obs})$. We assume uniform priors on $m_{\rm WDM}^{-1}$, so the final posterior is given by the product of two likelihoods. Figure 3 (left panel) shows the two likelihoods and the posterior obtained from combining these two likelihoods. Details of these calculations are presented in Appendix B.

The 90% quantile of the combined posterior is shown in the shaded region, yielding an upper limit $m_{\rm WDM}^{-1} < 0.035\ \mathrm{keV}^{-1}$, for $R = 5 \times 10^5\ \mathrm{yr}^{-1}$ and $T_{\rm obs} = 10$ yrs. Figure 3 (right panel) shows the upper limit of $m_{\rm WDM}^{-1}$ for different observation time periods and merger rates, including the Poisson uncertainties in the estimation.

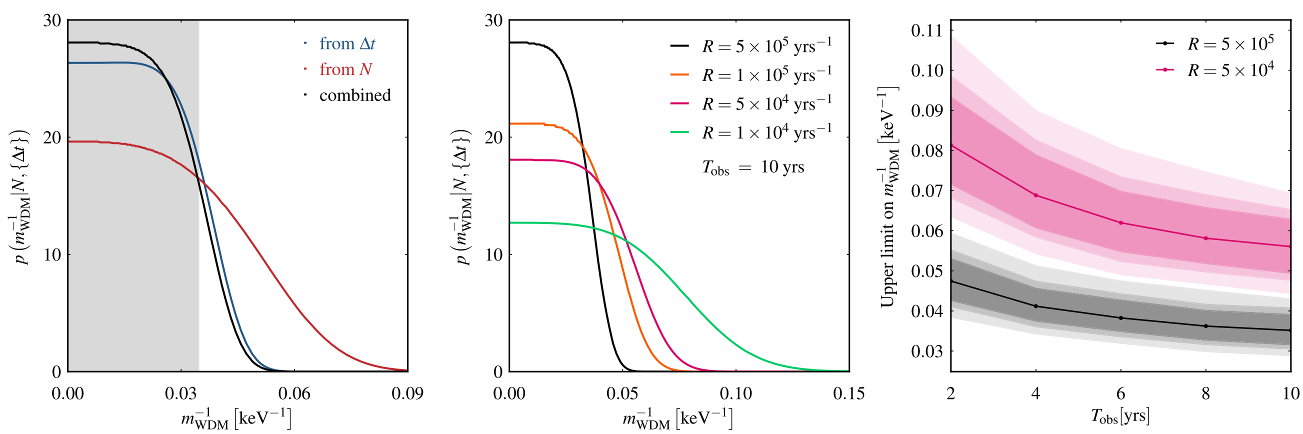

FIG. 3. Left panel: posterior distribution of $m_{\rm WDM}^{-1}$ computed from the number of lensed events and time delay distribution separately, along with the combined posterior. The simulated observations assume a BBH detection rate of $R = 5 \times 10^5\ \mathrm{yr}^{-1}$ and $T_{\rm obs} = 10$ yrs. The gray shaded region represents the 90% quantile of the combined posterior, yielding an upper limit of $m_{\rm WDM}^{-1} \leq 0.035$. Middle panel: combined constraints from GW lensing on $m_{\rm WDM}^{-1}$ assuming different detection rates (with $T_{\rm obs} = 10$ yrs). Right panel: 38%, 50%, and 68% credible intervals (denoted by different shades) of the distributions of the 90% upper limit of $m_{\rm WDM}^{-1}$ obtained from ~1000 simulated observations, each for different values of R and $T_{\rm obs}$.

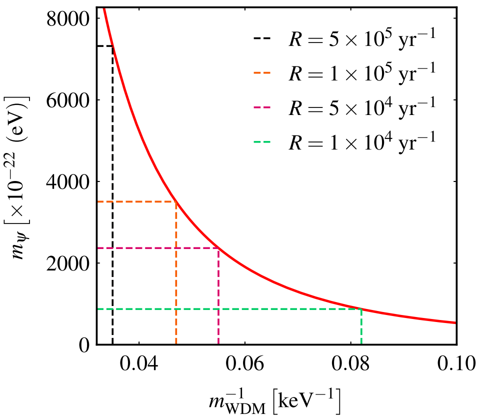

FIG. 4. Relation between m_ψ and $m_{\rm wdm}^{-1}$ obtained by equating the half mode length scale of FDM and WDM. The dashed vertical line shows the constraints on $m_{\rm wdm}^{-1}$ and corresponding dashed horizontal lines show the translated constraints on m_ψ. Different dashed lines are for different detection rates R, assuming $T_{\rm obs} = 10$ yrs.

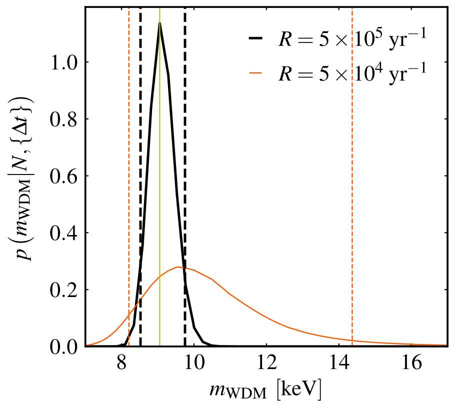

FIG. 5. Posterior distribution of $m_{\rm WDM}$ assuming that the true nature of the DM is described by the WDM model with $m_{\rm WDM} = 9$ keV (the yellow solid line in the middle). The plot shows the expected posteriors for $T_{\rm obs} = 10$ yrs assuming optimistic ($R = 5 \times 10^5$ yr^{-1}) and pessimistic ($R = 5 \times 10^4$ yr^{-1}) detection rates. Vertical dashed lines represent the 90% credible region of the posteriors.

The FDM also predicts a cutoff in the HMF at small scales, through a mechanism that is dependent on the de Broglie wavelength rather than a free-streaming length. We translate the constraints on the $m_{\rm WDM}$ to the mass of the FDM particle m_ψ (Fig. 4). For this, we simply equate the half mode length scale of both. We use the expression of the half mode length scale of FDM as given in [26], which considers a transfer function given in [17]. We calculate the half mode length scale for WDM using the HMFCALC package [69], which uses the formula given in [68] and a transfer function given in [67]. This is just a simple translation; in order to be more rigorous, one needs to use the HMF in FDM. Even though approximate, this gives us an idea of the prospective constraints on FDM using future observations of GW strong lensing.

We also check whether we will be able to measure the mass of the DM particle when it is actually warm. To check this, we simulate an observing scenario using the HMF of the WDM model with mass $m_{\rm WDM} = 9$ keV. Other details of the analysis are kept the same. As seen in Fig. 5, the true value of $m_{\rm WDM}$ is recovered within a 68% credible interval.

Outlook—Because of their inherent simplicity, GWs are unaffected by extinction, and selection effects in GW searches are well modeled. This makes GW strong lensing a cleaner probe of the nature of DM, as compared to, e.g., optical lensing. Constraints on the mass of the WDM-FDM particles expected from future GW strong lensing observations are better than the current bounds.

In our observables, there is no degeneracy between the DM properties and cosmological parameters. This should allow us to constrain both the nature of DM and cosmological parameters simultaneously from future observations. A wrong model of the HMF can bias our measurements. However, we should be able to identify the right HMF model from the data using Bayesian model selection. See Appendix C for a discussion on some of the potential systematic errors in our analysis.

A number of obstacles need to be addressed before this technique can be applied to real data. Properties of astrophysical sources and lenses determine both the number of lensed events and the distribution of their time delays. Some of the relevant parameters, such as the redshift distribution of BBH mergers, can be inferred from the large number of unlensed GW signals as well as the stochastic GW background. For other parameters, such as the distribution of the lens properties, will need to rely on large-scale cosmological simulations and galaxy surveys.

In this Letter, we have assumed that the lenses are modeled by simple SISs, whose parameters are obtained from the HMF using a simple prescription. We also neglected the halo substructure and the effect of baryons. Future analysis needs to model the lenses more accurately. We are working on improving this [78]. We neglected the selection function of the detectors assuming that XG detectors will observe all of the BBHs. In order to forecast the expected constraints from the upcoming upgrades of the current generation detectors [79–81], we will also need to take into account the selection effects, which are being studied [82]. There are other potential sources of systematics: the false positives produced by the methods used to identify strongly lensed signals can contaminate our observational sample and can bias the parameter inference. In [83] we demonstrated how we can model the contamination, thus evading systematic biases in the estimation of cosmological parameters from strongly lensed GW signals. A similar approach can be adopted here as well.

While we have focused on BBH signals, strongly lensed binary neutron star mergers, which are expected to be equally abundant [54,75,84], could also be used to probe cosmological parameters and the nature of DM. In contrast to BBH mergers which can be observed out to very large redshifts ($z \sim 10$), neutron star mergers can be observed out to smaller redshifts ($z \sim 1$) even using XG detectors. However, some of them would have an electromagnetic counterpart which will also be lensed. Such observations could allow us to probe the detailed profile of the lensing galaxy, potentially enabling us to put tighter constraints on the nature of DM. This is being explored in an ongoing work [85].

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End Matter

Appendix A: Modeling the expected number of lensed events and time delay distribution—Here we describe the modeling of the expected number of lensed events and the expected distribution of lensing time delay corresponding to different masses $m_{\rm WDM}$ for the WDM particle. We choose to parametrize these in terms of the inverse mass $m_{\rm WDM}^{-1}$ since it has a convenient lower bound of zero (the limit of CDM).

Expected number of lensed events:The expected number of lensed events, $\Lambda(m_{\rm WDM}^{-1})$, is given by Eq. (5). However, due to the finite observing time $T_{\rm obs}$, the number of observed events will be reduced to

$$\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs}) = \mathcal{S}(T_{\rm obs}, m_{\rm WDM}^{-1})\Lambda(m_{\rm WDM}^{-1}), \qquad (A1)$$

where $\mathcal{S}(T_{\rm obs}, m_{\rm WDM}^{-1})$ takes into account the selection effects introduced by the finite observation time:

$$\mathcal{S}(T_{\rm obs}, m_{\rm WDM}^{-1}) = \int_{\Delta t=0}^{T_{\rm obs}} p(\Delta t|m_{\rm WDM}^{-1}) \frac{(T_{\rm obs} - \Delta t)}{T_{\rm obs}} d\Delta t. \qquad (A2)$$

This selection function ensures that, for a time delay Δt, the first image must arrive at the detector within $T_{\rm obs} - \Delta t$ from the start of observation. Otherwise, the second image will arrive outside the observation window, rendering it undetectable.

Expected distribution of lensing time delays: We compute the expected time delay distribution $p(\Delta t|m_{\rm WDM}^{-1})$ by marginalizing the distribution of time delay over all other parameters $\vec{\lambda} \equiv \{y, \sigma, z_\ell, z_s\}$ that influence the time delay [see Eq. (7)]. Here, we discuss the choice of $p(\vec{\lambda}|m_{\rm WDM})$.

Assuming isotropy of space, the distribution of y becomes independent of both cosmological parameters and $m_{\rm WDM}^{-1}$, leading to

$$p(\vec{\lambda}|m_{\rm WDM}^{-1}) = p(y)p(\sigma, z_\ell, z_s|m_{\rm WDM}^{-1}), \qquad (A3)$$

where $p(y) \propto y$ with $y = (0, 1]$. The term $p(z_\ell, \sigma, z_s|m_{\rm WDM}^{-1})$ can be further split as

$$p(\sigma, z_\ell, z_s|m_{\rm WDM}^{-1}) = p(\sigma, z_\ell|z_s, m_{\rm WDM}^{-1})p_b(z_s), \qquad (A4)$$

where $p_b(z_s)$ is the expected measured distribution of source redshifts, while $p(\sigma, z_\ell|z_s, m_{\rm WDM}^{-1})$ is computed from the differential optical depth by [Eq. (11) in [83]]

$$p(\sigma, z_\ell|z_s, m_{\rm WDM}^{-1}) \propto \frac{d\tau}{dz_\ell d\sigma}(z_s, m_{\rm WDM}^{-1}). \qquad (A5)$$

This differential optical depth is derived from a HMF that depends on the nature of DM, making the time delay distribution sensitive to the value of $m_{\rm WDM}^{-1}$. We obtain the model time delay distribution $p(\Delta t|m_{\rm WDM}^{-1}, T_{\rm obs})$ from intrinsic time delay distribution $p(\Delta t|m_{\rm WDM}^{-1})$ after applying a selection condition that excludes time delays longer than the observation period $T_{\rm obs}$:

$$p(\Delta t|m_{\rm WDM}^{-1}, T_{\rm obs}) \propto p(\Delta t|m_{\rm WDM}^{-1}) \frac{(T_{\rm obs} - \Delta t)}{T_{\rm obs}} \Theta(T_{\rm obs} - \Delta t), \qquad (A6)$$

Appendix B: Bayesian inference—We assume that we have confidently detected N strongly lensed BBH events within an observation period of $T_{\rm obs}$, each producing two observable copies (lensed images). We aim to compute the posterior distribution of $m_{\rm WDM}^{-1}$, using the time delays $\{\Delta t_i\}_{i=1}^N$ from the N detected lensed events.

We can write the likelihood $p(N, \{\Delta t_i\}|m_{\rm WDM}^{-1}, T_{\rm obs})$ as a product of two likelihoods as N and $\{\Delta t_i\}$ are uncorrelated. The likelihood of observing N events can be described as a Poisson distribution:

$$p(N|m_{\rm WDM}^{-1}, T_{\rm obs}) = \frac{\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs})^N e^{-\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs})}}{N!}, \qquad (B1)$$

where $\Lambda(m_{\rm WDM}^{-1}, T_{\rm obs})$ is the expected number of lensed events for a given value of $m_{\rm WDM}^{-1}$.

The likelihood of observing a set of time delays $\{\Delta t_i\}$ can be written as

$$p(\{\Delta t_i\}|m_{\rm WDM}^{-1}, T_{\rm obs}) = \prod_{i=1}^{N} p(\Delta t_i|m_{\rm WDM}^{-1}, T_{\rm obs}). \qquad (B2)$$

Here, we assume that each BBH merger is an independent event and that the time delays are measured accurately and precisely; $p(\Delta t_i|m_{\rm WDM}^{-1}, T_{\rm obs})$ is the "model" time delay distribution, evaluated at the measured time delay Δt_i.

Appendix C: Systematic errors—We have investigated various sources of systematic errors in deriving constraints on the nature of DM using future observations of strongly lensed GW signals.

One potential source of systematic error is the HMF that is used to model the mass and redshift distribution of lenses. To get a sense of the systematic errors, we simulate the population of lenses using one HMF model (Behroozi [70]) and use another model (Jenkins [71]) for our inference, both implemented in the HMFCALC package [69]. Both of them are CDM HMF models, which are then converted to WDM models using the transfer function given in Eq. (3). As we see in the left panel of Fig. 6, the true value ($m_{\rm WDM}^{-1} = 0$) is not recovered if we use the wrong HMF model in our inference. This underlines the need for accurate models of the distribution of lens properties.

Though the true value of $m_{\rm WDM}$ is not recovered in the parameter inference, we perform a Bayesian model selection study using these two HMF models to calculate the Bayes factor (ratio of Bayesian evidence) between them. We repeat this exercise over a large number of random realizations of the same observing scenario and find that the Bayes factor overwhelmingly prefers the true HMF model (right panel of Fig. 6). Thus, if the correct HMF model is among the set of models that we consider for the parameter inference, we expect it to have the largest Bayesian evidence, thus helping us to evade systematic errors to a good extent.

We also check whether the change in the expected number of strongly lensed events and their time delay

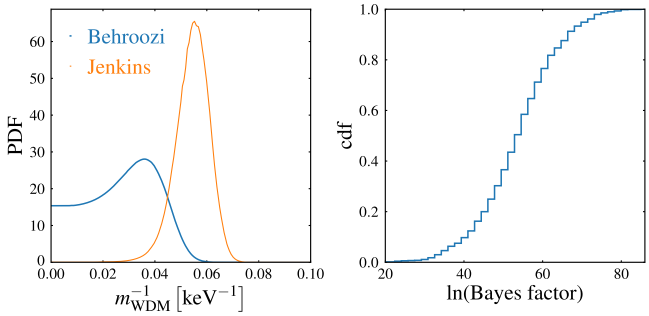

FIG. 6. Left panel: posterior distributions of m_{WDM}^{-1} computed from the number of lensed events and time delay distribution. The lensed events are simulated assuming a CDM HMF ($m_{\mathrm{WDM}}^{-1} \simeq 0$) described by the Behroozi model with merger rate $R = 5 \times 10^5\ \mathrm{yr}^{-1}$ and $T_{\mathrm{obs}} = 10$ yrs. The blue and orange lines show the posterior distribution of m_{WDM}^{-1} estimated using the Behroozi ("right") and Jenkins ("wrong") WDM HMF models, respectively. The true value of $m_{\mathrm{WDM}}^{-1} \simeq 0$ is not recovered when the wrong HMF model is employed in the inference. Right panel: cumulative distribution of the ratio of the evidence between the Behroozi and Jenkins models. This shows that the "right" model (Behroozi) is consistently preferred over the "wrong" model (Jenkins).

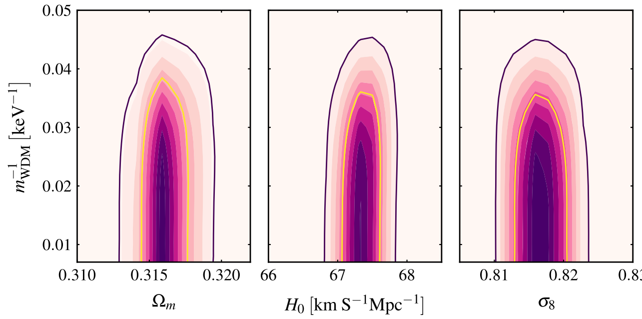

FIG. 7. Two-dimensional likelihoods of m_{WDM}^{-1}, Ω_M, H_0, and σ_8 that illustrate the (lack of) correlation between these parameters. Each plot shows a two-dimensional slice of the likelihood while keeping all other parameters to their true value. The colors represent the value of likelihood, and the contour lines show the 68% and 95% credible levels. It can be seen that there is no correlation between m_{WDM}^{-1} and the cosmological parameters.

distribution due to differences in the DM models could be mimicked by changes in cosmological parameters as studied in [49,83]. If such a degeneracy exists, the properties of the DM particle will be indistinguishable from cosmological parameters. To study this, we generate a sample of lensed BBH events assuming the standard value of the cosmological parameters that we used in the paper along with the CDM HMF. We then compute the likelihood by simultaneously varying m_{WDM} as well as the cosmological parameters H_0, Ω_M, and σ_8. Figure 7 shows various two-dimensional slices of the likelihood that illustrate the lack of correlation between m_{WDM} and the cosmological parameters. This is because WDM effects are only in the small scales (suppressing the abundance of low-mass halos and hence small time delays) while cosmological parameters affect all scales. This suggests that marginalizing over the cosmological parameters will not change the posterior of m_{WDM}^{-1} significantly.